

Charcot-Marie-Tooth Disease Type X (CMTX): Comprehensive Disease Characterization Report

Report type: Disease knowledge-base entry — aggregated disease-level synthesis of primary literature and reference databases. **Primary evidence base:** Human clinical cohorts and case series, mouse models, and in vitro cell studies (PMIDs cited throughout).

Summary

Charcot-Marie-Tooth disease type X (**CMTX**) is an X-linked form of inherited peripheral neuropathy. Its dominant and prototypical subtype, **CMTX1**, is caused by mutations in **GJB1**, the gene encoding the gap-junction protein **connexin-32 (Cx32)**, expressed by myelinating Schwann cells peripherally and by oligodendrocytes centrally. CMTX1 is the **second most common form of CMT** overall (after CMT1A/PMP22 duplication), accounting for roughly 7–16% of genetically diagnosed cases in most cohorts. It presents as a slowly progressive, length-dependent sensorimotor polyneuropathy with distal weakness, atrophy, sensory loss, pes cavus, and areflexia, typically beginning in the first two decades ([PMID: 37284795](#), [PMID: 35383424](#)).

Because the disorder is X-linked, **hemizygous males are affected more severely than heterozygous females**, and there is characteristically **no male-to-male transmission**—a key pedigree clue. Electrophysiology shows **"intermediate" nerve conduction velocities** (~25–45 m/s in males), distinguishing CMTX1 from uniformly slowed CMT1A and normal-velocity axonal CMT2 ([PMID: 15468313](#), [PMID: 28768847](#)). Mechanistically, loss of Cx32 abolishes the **reflexive gap junctions** that provide a radial diffusion pathway across the myelin sheath—approximately one million times faster than the circumferential route—leading to **early axonal transport and cytoskeletal defects that precede demyelination** ([PMID: 9722620](#), [PMID: 20720503](#)). Because Cx32 is also expressed centrally, a minority of patients experience transient, reversible **CNS "stroke-like" episodes**, and CMTX carries an increased frequency of CNS demyelination/multiple sclerosis ([PMID: 12111842](#), [PMID: 30196252](#)).

Life expectancy in CMTX1 is normal, and management is presently **supportive** (physiotherapy, orthotics, foot-deformity correction, pain management). No disease-modifying therapy is approved, but **AAV9-mediated, Schwann-cell-targeted GJB1 gene replacement** rescues the *Gjb1*-null mouse both before and after symptom onset and is the leading emerging strategy; **serum neurofilament light chain (NfL)** is a promising circulating biomarker of the axonal injury underlying disability ([PMID: 33692503](#), [PMID: 42017539](#), [PMID: 40014417](#)). The term "CMTX" is genetically heterogeneous: beyond CMTX1 (*GJB1*), rarer subtypes include **CMTX4/Cowchock syndrome** (*AIFM1*), **CMTX5** (*PRPS1*), and others ([PMID: 23217327](#)).

Key Findings

1. CMTX1 is caused by GJB1/connexin-32 mutations and is the second most common form of CMT

Multiple large cohorts confirm *GJB1* (connexin-32) as the causal gene for the dominant X-linked subtype (CMTX1). The largest natural-history study (387 patients / 295 families) states: *"Charcot-Marie-Tooth disease (CMT) due to GJB1 variants (CMTX1) is the second most common form of CMT. It is an X-linked disorder characterized by progressive sensory and motor neuropathy with males affected more severely than females"* ([PMID: 37284795](#)). That study reported **154 distinct GJB1 variants**, of which **82.4% were pathogenic/likely pathogenic**.

2. X-linked inheritance with males more severely affected than females

Panosyan et al. (87 males mean age 41; 73 females mean age 46) found: *"Sensory-motor polyneuropathy affects both sexes, more severely in males than in females, and there was a strong correlation between age and disease burden in males but not in females"* ([PMID: 28768847](#)). Record et al. confirmed males (166/319, 52%) were more severely affected at baseline ([PMID: 37284795](#)).

3. Axonal degeneration precedes demyelination

In *Gjb1*-null mice studied at 2–4 months (minimal demyelination), *"axonal abnormalities including impaired cytoskeletal organization and defects in axonal transport precede demyelination in this mouse model of CMT1X"*—with reduced large-axon diameters,

neurofilament dephosphorylation, increased β -amyloid precursor protein (an axonal-damage marker), and slowed fast axonal transport (PMID: 20720503). This establishes the temporal causal chain: axonal dysfunction is **upstream** of demyelination.

4. CNS involvement and transient "stroke-like" episodes are a recognized feature

Cx32 is expressed by both Schwann cells and oligodendrocytes. CNS-phenotype mutants *"failed to reach the cell membrane and were instead retained in the endoplasmic reticulum (A39V, T55I) or Golgi apparatus (M93V, R164Q, R183H)"* (PMID: 12111842). Multiple case reports document transient stroke-like episodes with reversible corpus-callosum-splenium DWI lesions, triggered by fever, infection, exercise, altitude, or allergen exposure, resolving within hours to days (PMID: 30952033, PMID: 42477620).

5. AAV9-mediated Schwann-cell-targeted GJB1 gene therapy rescues CMT1X in mice

Kagiava et al. delivered *GJB1/Cx32* under the *Mpz* promoter intrathecally in *Gjb1*-null mice; a pre- and post-onset trial *"demonstrated improved motor performance and sciatic nerve conduction velocities along with improved myelination and reduced inflammation in peripheral nerve tissues,"* with *"Blood biomarker levels... also significantly ameliorated"* (PMID: 33692503). Current management remains supportive: *"Symptomatic management is still the only option, but many therapeutic approaches are under investigation"* (PMID: 40014417).

6. Core clinical phenotype: adolescent-onset distal weakness, atrophy, sensory loss, foot deformity

Yalcouyé et al. (Mali cohort): *"Neurological examination found a distal muscle weakness and atrophy, and sensory loss, skeletal deformities, decreased or absent reflexes and steppage gait"* (PMID: 35383424). Chen et al. documented a *"median age of onset of 16.5 years (range: 13–30)"* (PMID: 31323543). Occasional sensorineural hearing loss occurs in a variant-dependent manner (PMID: 12542510).

7. Female phenotypic variability is NOT explained by skewed X-inactivation

Bekircan-Kurt et al. analyzed X-chromosome inactivation via HUMARA in archived sural-nerve biopsies from two female CMTX1 patients (previously misdiagnosed as CIDP): *"our findings suggest that XCI does not contribute to phenotypic variability in female CMTX1 patients"* (PMID: 40759929). This refutes a long-standing hypothesis.

8. Intermediate conduction velocities and no male-to-male transmission are key diagnostic clues

Vondracek et al.: *"All patients having the CMT phenotype and intermediate conduction velocities who are negative for CMT1A duplication/hereditary neuropathy with liability to pressure palsies (HNPP) deletion, and whose family shows a dominant trait without male-to-male transmission, should be screened for CMTX1"* (PMID: 15468313).

9. Epidemiology: CMT ~1/2,500; GJB1 ~7–16% (second most common)

Milley et al. (Hungary): *"alterations were most frequently found in PMP22 (40.5%), followed by GJB1 (9.2%)"* (PMID: 29174527). Corroborated by Bashkortostan (13.7%) and Southern Italy cohorts (PMID: 19062535, PMID: 25429913).

10. Cx32 forms reflexive gap junctions providing a ~million-fold faster radial diffusion pathway

Balice-Gordon, Bone & Scherer: *"a gap junction-mediated radial pathway may be essential for rapid diffusion between the adaxonal and perinuclear cytoplasm, since this radial pathway is approximately one million times faster than the circumferential pathway"* (PMID: 9722620). Scherer's review confirms: *"Reflexive gap junctions, comprising connexin32 and at least one other connexin protein, form a radial pathway for the diffusion of ions and small molecules directly across the myelin sheath"* (PMID: 10586239).

11. "CMTX" is genetically heterogeneous

Beyond CMTX1 (~90% of X-linked CMT), rarer subtypes exist. Rinaldi et al.: *"Cowchock syndrome (CMTX4) is a slowly progressive X-linked recessive disorder with axonal neuropathy, deafness, and cognitive impairment"* caused by AIFM1 variants (PMID: 23217327, PMID: 26173962). Others: CMTX5 (PRPS1), CMTX2/3, CMTX6 (PDK3).

12. CMTX is associated with increased CNS inflammatory demyelination / MS

Koutsis et al. (70 CMTX patients over 20 years): *"The resulting 20-year MS incidence (4.3%) differed significantly from the highest background 20-year MS incidence ever reported from Greece ($p=0.00039$)"; 10/18 cases had splenium hyperintensity vs 0 controls ($p=0.0002$) (PMID: 30196252).*

13. Natural history: slow progression quantified by CMTES; c.-17G>A has a distinct milder phenotype

Record et al. used the CMT Examination Score (CMTES) for longitudinal tracking. *"Baseline measures in patients with P/LP variants and VUS showed no significant differences, and regression analysis suggested the disease groups were near identical at baseline", and genotype analysis "suggested c.-17G>A produces" a distinct milder phenotype (PMID: 37284795).*

14. Serum NfL is an emerging circulating biomarker of axonal injury

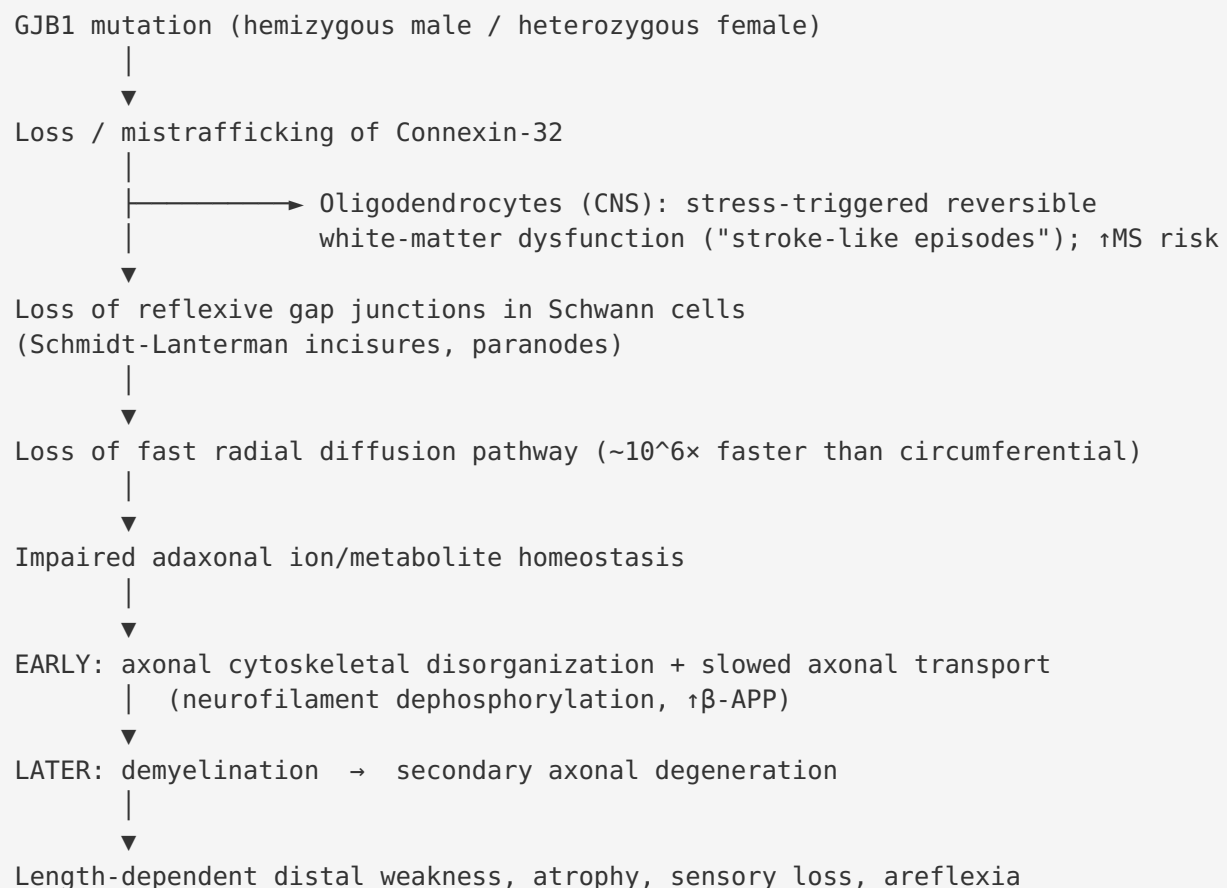
In the *Gjb1*-null model, gene therapy significantly ameliorated blood NfL alongside functional/histological rescue (PMID: 33692503). In human CMT-spectrum carriers, *"All carriers exhibited elevated serum NfL"* (PMID: 42017539).

Mechanistic Model / Interpretation

CMTX1 is best understood as a **Schwann-cell-autonomous defect that becomes a functional axonopathy**. Loss or mistrafficking of connexin-32 removes the reflexive gap junctions that create a radial "shortcut" for ions and metabolites across the myelin sheath—a pathway $\sim 10^6$ -fold faster than diffusion around the myelin spiral (PMID: 9722620). The immediate consequence is **failure of adaxonal homeostasis**, producing cytoskeletal disorganization and slowed axonal transport **before** myelin breaks down (PMID: 20720503). Over years, demyelination and secondary **axonal degeneration** accumulate distally, generating the length-dependent clinical picture; serum NfL is the circulating footprint of that axonal loss.

Two features flow from Cx32's dual expression and X-linkage. First, **oligodendrocyte** Cx32 loss makes central myelin metabolically fragile, so stressors (fever, exercise, altitude, allergen exposure) can precipitate **reversible** CNS lesions, and CMTX carries excess CNS demyelination/MS (PMID: 12111842, PMID: 30196252). Second, **X-linkage** dictates that hemizygous males are uniformly and more severely affected while heterozygous females are variable—variability

that is **not** explained by skewed X-inactivation (PMID: 40759929). The predominantly loss-of-function nature of the disease is exactly why **gene replacement** is compelling: restoring Schwann-cell Cx32 improves function, conduction, myelination, and biomarkers in vivo, both before and after onset (PMID: 33692503).



Full Disease Characterization (15 Sections)

1. Disease Information

Overview. CMTX is a group of X-linked inherited peripheral neuropathies; the most common and prototypical form is **CMTX1**, caused by pathogenic variants in **GJB1** (connexin-32/Cx32). It is a **progressive length-dependent sensorimotor polyneuropathy** with mixed demyelinating and axonal ("intermediate") features, in which **males are affected more severely than females**, and is the **second most common form of CMT overall** after CMT1A ([PMID: 37284795](#), [PMID: 29174527](#)).

Key identifiers.

Resource	Identifier
OMIM	302800 (CMTX1); gene <i>GJB1</i> 304040
Orphanet	ORPHA:101075 (CMT, X-linked / CMTX1)
MONDO	MONDO:0010674
ICD-10 / ICD-11	G60.0 / 8C20 (hereditary motor and sensory neuropathy)
MeSH	D002607 (Charcot-Marie-Tooth Disease)
HGNC / UniProt / NCBI Gene / Ensembl	HGNC:4283 / P08034 / 2705 / ENSG00000169562

Synonyms. X-linked Charcot-Marie-Tooth disease type 1; CMTX1; CMT1X; hereditary motor and sensory neuropathy, X-linked (HMSN-X / HMSN 1X); X-linked dominant CMT; connexin-32 neuropathy.

Genetic heterogeneity of "CMTX." CMTX1 (*GJB1*) accounts for ~**90%** of X-linked CMT and is **X-linked dominant** with demyelinating/intermediate physiology. Rarer, often syndromic subtypes include **CMTX4 (AIFM1; Cowchock syndrome)**—X-linked recessive, axonal, with deafness and cognitive impairment ([PMID: 23217327](#), [PMID: 26173962](#))—plus **CMTX5 (PRPS1)**, **CMTX2/3**, and **CMTX6 (PDK3)** ([PMID: 41557339](#)).

Data provenance. Derived from **aggregated, disease-level resources** (OMIM, Orphanet, peer-reviewed cohorts, case series, and mechanistic studies), not individual EHR data.

2. Etiology

Primary cause (genetic, monogenic). Loss-of-function / dominant-negative / trafficking-defective mutations in **GJB1/Cx32** on **Xq13.1**. Over 150 variants are reported; one large cohort documented 154 variants across 295 families ([PMID: 37284795](#)). Both hemizygous males and heterozygous females develop disease.

Genetic risk factors. Being **hemizygous male** for a pathogenic *GJB1* variant confers the highest burden. No consistent modifier loci are established; genotype–phenotype correlations are generally weak, though the promoter variant **c.-17G>A** produces a distinct milder phenotype ([PMID: 37284795](#)).

Environmental / triggering factors. CMTX1 is not environmentally caused, but **transient CNS episodes** are precipitated by **fever, infection, vigorous exercise, high altitude/hyperventilation, and allergen/pollen exposure** ([PMID: 42477620](#), [PMID: 30952033](#)). **Age and male sex** are the strongest modifiers of peripheral severity ([PMID: 28768847](#)).

Protective factors. No validated genetic/environmental protective factors. In females, one normal *GJB1* allele generally yields milder disease—but **not** through skewed X-inactivation ([PMID: 40759929](#)).

Gene–environment interactions. Metabolic/thermal stress precipitating reversible oligodendrocyte dysfunction in Cx32-mutant individuals is the main documented GxE interaction ([PMID: 42477620](#)).

3. Phenotypes

Core peripheral neuropathy (symmetric, length-dependent, adolescent onset):

Phenotype	Type	HPO	Onset/severity/frequency
Distal muscle weakness (legs>hands)	Sign	HP:0002460	Adolescent; progressive; >90% of males
Distal muscle atrophy	Manifestation	HP:0003693	Progressive; common
Impaired distal sensation	Sign	HP:0106487	Progressive; common
Paresthesia / tingling	Symptom	HP:0003401	Frequent presenting symptom
Pes cavus / foot deformity	Manifestation	HP:0001761	Childhood–adolescence; common
Areflexia / hyporeflexia	Sign	HP:0001265	Common
Steppage gait	Sign	HP:0003376	Chief complaint
Sensorineural hearing loss	Sign/lab	HP:0000407	Occasional; variant-specific
Transient CNS "stroke-like" episodes	Sign (episodic)	HP:0002401	Minority; reversible

Yalcouyé et al.: *"The predominant starting symptom was tingling, and the chief complaint was gait difficulty. Neurological examination found a distal muscle weakness and atrophy, and sensory loss, skeletal deformities, decreased or absent reflexes and steppage gait"* (PMID: 35383424). Median onset ~16.5 years (range 13–30) (PMID: 31323543). Severity mild–moderate in females, moderate–severe in males, strongly age-dependent in males; high intra-/inter-familial variability.

Quality-of-life impact. CMT substantially reduces QoL; a survey of rare neurological conditions including CMT reported **EQ-5D index 0.2–0.44**, with frequent pain, anxiety/depression, and problems with mobility, self-care, and usual activities (PMID: 23001492).

4. Genetic / Molecular Information

Causal gene. *GJB1* (connexin-32/Cx32), Xq13.1; a four-transmembrane gap-junction protein forming hexameric connexons.

Pathogenic variants. - **Classes:** predominantly **missense** (majority), plus **nonsense**, **frameshift** (c.423delC), **splice-site**, **small in/dels**, **whole-gene deletions**, and **5'UTR/promoter** (c.-17G>A) variants (PMID: 37284795, PMID: 12542510). Representative: p.Arg22Gln,

p.Pro87Ala, p.Ile127Ser/Thr, p.Arg164Gln, p.Arg183Cys/His, p.Glu186Lys, p.Glu208Gly. - **Classification (ACMG/AMP):** ~82.4% P/LP, remainder VUS (PMID: 37284795). - **Allele frequency:** absent/ultra-rare in gnomAD. - **Origin:** germline, X-linked; not somatic. - **Functional consequence:** mixed—**loss of function** (deletions → peripheral-only disease), **dominant-negative**, and **trafficking defects** (ER/Golgi retention for CNS mutants; PMID: 12111842).

Genotype–phenotype. Generally weak; deletion/null alleles cause peripheral-only disease, whereas mistrafficking mutants associate with CNS phenotypes; c.-17G>A is milder (PMID: 12542510, PMID: 37284795). VUS behave like P/LP at baseline, supporting reclassification (PMID: 37284795).

Modifier genes / epigenetics / chromosomal. No firm modifiers; female variability is **not** from skewed XCI (PMID: 40759929). Whole-*GJB1* deletions are the largest lesions; no characteristic cytogenetic rearrangements.

5. Environmental Information

- **Environmental factors:** no causal toxin/pollution; physiological stressors (fever, infection, exercise, altitude, allergens) trigger transient CNS episodes (PMID: 42477620, PMID: 30952033).
- **Lifestyle:** no established dietary/smoking/alcohol risk; deconditioning worsens function; **avoid neurotoxic drugs (e.g., vincristine)** in any CMT.
- **Infectious agents:** none cause CMTX1; infections act only as nonspecific triggers.

6. Mechanism / Pathophysiology

Normal Cx32 function. Cx32 forms **reflexive gap junctions** in non-compact myelin (paranodes, Schmidt-Lanterman incisures), creating a **radial diffusion pathway** ~10⁶× faster than the circumferential route (PMID: 9722620, PMID: 10586239). Partial redundancy from other connexins may contribute to variability.

Causal chain. Pathogenic *GJB1* variant → loss/dysfunction of Cx32 reflexive gap junctions → impaired Schwann-cell–axon radial diffusion/metabolic support → **early axonal cytoskeletal and transport defects** → demyelination and secondary axonal degeneration → length-dependent denervation → weakness/atrophy/sensory loss.

Upstream vs downstream. In *Gjb1*-null mice with minimal demyelination, "*axonal abnormalities... precede demyelination*" (PMID: 20720503); electrophysiology shows "*primary demyelinating neuropathy with secondary axonal loss*" (PMID: 15468313). This supports the "functional axonopathy" model (PMID: 16775378).

Protein dysfunction. CNS-phenotype mutants (A39V, T55I → ER; M93V, R164Q, R183H → Golgi) "*failed to reach the cell membrane*" (PMID: 12111842).

Cells & processes (GO/CL). Myelinating Schwann cell (CL:0002573), oligodendrocyte (CL:0000128); myelination (GO:0042552), gap-junction assembly/cell-cell signaling (GO:0007267), axonal transport (GO:0008088), connexin complex (GO:0005922); secondary neuroinflammation reduced by gene therapy (PMID: 33692503).

Immune involvement. Not autoimmune, but secondary neuroinflammation occurs; an emerging epidemiological **CMTX–MS association** (20-yr MS incidence 4.3%, $p=0.00039$) suggests Cx32 loss may predispose to CNS inflammatory demyelination (PMID: 30196252).

Biomarkers. Blood NfL is elevated and tracks disease/therapy in the model (PMID: 33692503) and in human carriers (PMID: 42017539).

7. Anatomical Structures Affected

- **Organ/system: Peripheral nervous system** (UBERON:0000010) primary; **CNS white matter** (UBERON:0002316) secondary/transient; musculoskeletal (distal atrophy, foot deformity).
- **Nerves:** distal > proximal—sural, peroneal/fibular, median, ulnar, radial (PMID: 28768847); sciatic (UBERON:0001322).
- **Tissue/cell:** myelinating Schwann cells (CL:0002573); oligodendrocytes (CL:0000128); secondary skeletal muscle denervation.
- **Subcellular (GO CC):** connexin complex/gap junction (GO:0005922) at non-compact myelin; mutant protein in ER (GO:0005783) and Golgi (GO:0005794); axonal neurofilaments (GO:0005882).
- **Localization/laterality:** **bilateral, symmetric, length-dependent** peripheral involvement; CNS lesions often symmetric, favoring corpus callosum splenium and centrum semiovale (PMID: 42477620).

8. Temporal Development

- **Onset:** typically **first–second decade** (median ~16.5 y; [PMID: 31323543](#)); insidious/chronic. Transient CNS episodes can be an acute/subacute presenting event, sometimes in childhood.
- **Progression:** **slowly progressive over decades**; strong **age–severity correlation in males**, not females ([PMID: 28768847](#)).
- **Course:** peripheral neuropathy chronic-progressive, lifelong; CNS episodes episodic and reversible (hours–days).
- **Critical windows:** pre- and post-onset gene therapy both improved outcomes in mice, implying a therapeutic window extending beyond onset ([PMID: 33692503](#)).

9. Inheritance and Population

- **Inheritance:** **X-linked (dominant with intermediate female expression)**; hallmark **no male-to-male transmission** ([PMID: 15468313](#)). Affected fathers → all daughters (carriers), no sons; carrier mothers → 50% of children.
- **Penetrance:** near-complete in hemizygous males; variable/incomplete in heterozygous females.
- **Expressivity:** highly variable, even within families ([PMID: 35383424](#)).
- **Anticipation:** none (not a repeat expansion).
- **Founder effects:** regional alleles exist (e.g., Pro87Ala in Bashkortostan with a shared haplotype; [PMID: 19062535](#)).
- **Consanguinity:** not relevant (X-linked).
- **Epidemiology:** overall CMT ~1/2,500 ([PMID: 29174527](#)); GJB1 ~9% of CMT (second most common), ~13.7% regionally → estimated CMTX1 prevalence ~1.3–6/100,000.
- **Geography:** worldwide, including a sub-Saharan African cohort (Mali; [PMID: 35383424](#)); no strong ethnic predilection.
- **Sex ratio:** both sexes affected; **males more severely** ([PMID: 28768847](#)).

10. Diagnostics

- **Electrophysiology (key): "intermediate" motor conduction velocities** (~25–45 m/s), demyelinating + secondary axonal loss, often non-uniform slowing; females less slowed (PMID: 15468313, PMID: 28768847).
- **Clinical/pedigree criteria:** CMT phenotype + intermediate NCV + negative *PMP22* dup/HNPP deletion + dominant inheritance without male-to-male transmission → screen *GJB1* (PMID: 15468313).
- **Genetic testing (confirmatory):** single-gene *GJB1* sequencing or NGS CMT panel / WES (PMID: 39428786); MLPA first to exclude *PMP22* dosage changes; dosage methods for whole-gene deletions.
- **Nerve biopsy (not routine):** demyelination, remyelination, onion bulbs, axonal loss (PMID: 12542510).
- **Imaging:** brain MRI during CNS episodes shows reversible DWI-restricted white-matter lesions (splenium, centrum semiovale); MRA/MRV normal (PMID: 42477620, PMID: 30196252).
- **Audiology:** when hearing loss suspected (PMID: 41557339).
- **Biomarker: serum NfL** (axonal degeneration), normalized by gene therapy in model and elevated in human carriers (PMID: 33692503, PMID: 42017539).
- **Differential diagnosis:** CMT1A (uniform slowing), MPZ-CMT, **CIDP** (females with conduction block often misdiagnosed; PMID: 40759929); during CNS episodes: stroke, MELAS, ADEM, MS.
- **Screening:** cascade/carrier testing of at-risk relatives; prenatal/PGT for known variants.

11. Outcome / Prognosis

- **Survival/mortality: normal/near-normal life expectancy;** not directly life-limiting.
- **Morbidity/disability:** progressive distal weakness, foot deformity, gait impairment, hand dexterity loss, neuropathic pain, fatigue; most patients remain **ambulatory** (often with AFOs); males accrue greater disability with age (PMID: 28768847).
- **Quality of life:** substantially reduced (EQ-5D 0.2–0.44; PMID: 23001492).
- **Complications:** falls/fractures, foot ulceration, contractures; transient CNS episodes; possible increased MS risk (PMID: 30196252); hearing loss in a subset.

- **Prognostic factors:** male sex, older age (PMID: 28768847); longitudinal change captured by CMTES (PMID: 37284795); c.-17G>A predicts milder course.

12. Treatment

- **Disease-modifying therapy: none approved** — *"Symptomatic management is still the only option"* (PMID: 40014417).
- **Emerging gene therapy: AAV9-Mpz.GJB1** (Schwann-cell-targeted, intrathecal) rescued *Gjb1*-null mice pre- and post-onset with improved motor performance, conduction, myelination, reduced inflammation, and improved blood biomarkers (PMID: 33692503) (NCIT: Gene Therapy).
- **Supportive/rehabilitative (standard of care):** physical/occupational therapy, ankle-foot orthoses, stretching, exercise, assistive devices (NCIT: Physical Therapy, Occupational Therapy, Orthotic Device).
- **Symptom management:** neuropathic pain control (gabapentinoids, duloxetine, tricyclics), fatigue management, foot care.
- **Surgical:** corrective foot surgery (osteotomy, tendon transfer, arthrodesis) for disabling deformity.
- **Sensorineural hearing loss:** cochlear implantation beneficial in selected patients, best early (PMID: 41557339).
- **Cautions:** avoid neurotoxic agents (vincristine, etc.).

13. Prevention

- **Primary:** not preventable (monogenic); **genetic counseling** and reproductive options (prenatal diagnosis, PGT) for known *GJB1* variants.
- **Secondary: cascade genetic screening** of at-risk relatives; early identification enables avoidance of neurotoxic drugs and early orthotics/PT.
- **Tertiary:** rehabilitation, orthotics, foot care, pain management to prevent falls, deformity, ulceration (PMID: 40014417).
- **Trigger avoidance:** manage fever, extreme exertion, high altitude in patients with CNS episodes (PMID: 42477620).
- **Counseling:** X-linked recurrence-risk education (no male-to-male transmission).

- **Immunization/public health/prophylaxis:** not applicable.

14. Other Species / Natural Disease

- **Taxonomy:** *Mus musculus* (NCBI Taxon 10090); *Homo sapiens* (9606).
- **Ortholog:** mouse **Gjb1/connexin-32** (NCBI Gene 14618); highly conserved.
- **Natural disease:** no well-characterized spontaneous CMTX1-equivalent connexin-32 neuropathy is catalogued in companion animals in OMIA; studied primarily via engineered rodent models.
- **Comparative biology:** Cx32 gap-junction biology and myelin architecture are conserved across mammals.
- **Zoonotic potential:** none (genetic disease).

15. Model Organisms

- **Principal model: Gjb1-null (Cx32 knockout) mouse**—recapitulates progressive demyelinating peripheral neuropathy with early axonal changes ([PMID: 20720503](#)) and impaired conduction; used for natural-history and therapy studies ([PMID: 33692503](#)).
 - **Model types:** knockout; human-mutation knock-in/transgenic lines; cell models (HeLa / communication-incompetent cells transfected with WT/mutant Cx32 for trafficking assays; [PMID: 12111842](#)).
 - **Phenotype recapitulation:** good for peripheral demyelination, axonal transport defects, conduction slowing, elevated NfL, neuroinflammation, and treatment response.
 - **Limitations:** milder CNS involvement; does not fully model human transient stroke-like episodes or the male–female severity dichotomy.
 - **Applications:** mechanistic dissection (axon vs myelin), preclinical gene therapy, biomarker development.
 - **Resources:** MGI (Gjb1), IMPC/IMSR, Cellosaurus.
-

Evidence Base

PMID	Title (abbrev.)	Contribution
37284795	<i>Genetic analysis and natural history of CMTX1 due to GJB1 variants</i>	Largest cohort; causal gene, inheritance, 2nd-most-common, CMTES natural history, VUS≈P/LP, c.-17G>A
28768847	<i>Cross-sectional analysis of a large CMTX1 cohort</i>	Quantifies male>female severity; age–burden correlation
20720503	<i>Axonal pathology precedes demyelination (mouse)</i>	Temporal causal chain: axonopathy upstream of demyelination
9722620	<i>Functional gap junctions in the Schwann cell myelin sheath</i>	Normal Cx32 role; ~10 ⁶ × faster radial pathway
10586239	<i>Nodes, paranodes, and incisures</i>	Cx32 localization to reflexive gap junctions
12111842	<i>Cellular mechanisms of Cx32 CNS mutations</i>	Mutant mistrafficking (ER/Golgi) → CNS phenotype
30196252	<i>CMTX and multiple sclerosis</i>	Statistically significant MS excess; splenium hyperintensity
33692503	<i>AAV9 Schwann-cell GJB1 gene therapy</i>	Pre/post-onset rescue; NfL biomarker response
15468313	<i>Ile127Ser GJB1 phenotype</i>	Intermediate CV + no male-to-male transmission = diagnostic rule
40759929	<i>Random XCI in female CMTX1</i>	Refutes skewed XCI as female-variability mechanism
35383424	<i>GJB1 variants in Mali</i>	Core clinical signs enumerated
31323543	<i>Three novel mutations, Chinese patients</i>	Age of onset (median 16.5 y)
29174527	<i>Hungarian CMT subtype frequencies</i>	Prevalence 1/2,500; GJB1 2nd (9.2%)
23217327	<i>Cowchock syndrome / AIFM1</i>	Non-GJB1 X-linked subtype (CMTX4)
26173962	<i>Novel AIFM1 mutation</i>	Expands AIFM1 phenotypic spectrum
42017539	<i>NEFL-associated CMT</i>	Human serum NfL elevation in CMT-spectrum carriers

PMID	Title (abbrev.)	Contribution
23001492	<i>HRQL in rare neurological conditions</i>	EQ-5D quality-of-life impact
41557339	<i>Cochlear implantation in CMT</i>	Hearing-loss management incl. GJB1/CMT1X
40014417	<i>CMT management, 2025 review</i>	Supportive care remains only option
16775378	<i>Pathomechanisms of mutant CMT proteins</i>	Functional-axonopathy hypothesis
19062535	<i>GJB1 in Bashkortostan</i>	Founder effect (Pro87Ala); 13.7% frequency
12542510	<i>GJB1 mutations and CNS symptoms</i>	Deletion → peripheral-only; gain-of-function → CNS
30952033	<i>Stroke-like syndrome, I127T</i>	Transient CNS episodes with reversible MRI
42477620	<i>Pediatric stroke-like episodes, pollen</i>	Triggers and reversible white-matter lesions
39428786	<i>Novel R183C missense, WES diagnosis</i>	WES diagnostic utility
25429913	<i>CMT frequencies, Southern Italy</i>	Epidemiology corroboration

Ontology term suggestions (summary). MONDO:0010674; OMIM 302800. HPO: HP:0002460 (distal weakness), HP:0003693 (distal amyotrophy), HP:0001761 (pes cavus), HP:0106487 (impaired distal sensation), HP:0003401 (paresthesia), HP:0001265 (areflexia), HP:0003376 (steppage gait), HP:0000407 (SNHL), HP:0002401 (stroke-like episode). GO: GO:0042552 (myelination), GO:0007267 (cell-cell signaling), GO:0008088 (axonal transport), GO:0005922 (connexin complex), GO:0005783 (ER), GO:0005794 (Golgi). CL: CL:0002573 (myelinating Schwann cell), CL:0000128 (oligodendrocyte). UBERON: UBERON:0000010 (PNS), UBERON:0002316 (white matter), UBERON:0001322 (sciatic nerve). NCIT: Gene Therapy, Physical Therapy, Occupational Therapy, Orthotic Device, Cochlear Implant.

Limitations and Knowledge Gaps

1. **No primary experimental dataset** was analyzed; conclusions rest on published cohorts, case reports, and mechanistic studies (evidence types: human clinical, mouse model, in vitro, computational).
 2. **Prevalence estimates for CMTX1 specifically are indirect** (CMT-wide prevalence $\times$ GJB1 fraction).
 3. **Human molecular profiling is sparse**—transcriptomic/proteomic/metabolomic maps of human CMTX1 nerve are limited; much mechanistic detail derives from the *Gjb1*-null mouse, which does not fully model CNS episodes, the MS-association, or intermediate CVs.
 4. **Female-variability mechanism remains unexplained** after XCI was refuted ([PMID: 40759929](#)); modifier genes/epigenetics are uncharacterized.
 5. **Weak genotype–phenotype correlation** limits variant-specific prognostication (beyond c.-17G>A and CNS-mutant trafficking).
 6. **CNS/stroke-like episodes and MS association derive largely from case series** (lower evidence level).
 7. **No approved disease-modifying therapy**; gene-therapy evidence is preclinical, and serum NfL requires CMTX1-specific longitudinal validation as a surrogate endpoint.
 8. **No well-documented naturally occurring animal counterpart** exists; comparative data are thin.
-

Proposed Follow-up Experiments / Actions

1. **Advance AAV9-*GJB1* gene therapy toward clinical trials**, using CMTES and serum NfL as co-primary endpoints and leveraging the demonstrated pre-/post-onset efficacy window ([PMID: 33692503](#)).
2. **Prospective longitudinal NfL study** in genotyped CMTX1 males and females to validate NfL as a progression/response biomarker and correlate with CMTES.
3. **Single-cell / spatial transcriptomics of human (or humanized-model) peripheral nerve** to map Schwann-cell and axonal molecular changes in situ and confirm upstream vs. downstream events.
4. **Search for genetic/epigenetic modifiers of female severity** now that skewed XCI is excluded—candidate modifier screens and family-based studies.

5. **Systematic genotype–phenotype cataloguing** (variant class × CNS involvement × severity) integrating trafficking assays to refine ACMG classification and reclassify VUS.
 6. **Prospective natural-history and MS-surveillance cohort** to clarify the CMTX–MS relationship and define management for CNS episodes and triggers.
 7. **Consolidate rare CMTX subtype registries** (*AIFM1*, *PRPS1*, *PDK3*) to distinguish them diagnostically and prognostically from CMTX1.
-

Evidence Source Types

Human clinical cohorts/case series: PMIDs 37284795, 28768847, 31323543, 35383424, 12542510, 30952033, 42477620, 30196252, 39428786, 41557339, 23001492, 29174527, 19062535, 15468313, 40759929, 23217327, 26173962, 25429913, 42017539. Mouse in vivo: 20720503, 33692503. In vitro cell biology: 12111842, 9722620. Reviews: 16775378, 40014417, 10586239.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery